CNN-Based Water Pollution Image Classification for Environmental Monitoring: A Comparative and Honestly Reported Study of a Custom Convolutional Network and a MobileNetV2 Transfer-Learning Baseline on a Small Image Dataset

Dharmik Bhisara

Department of Data Science, University of Europe for Applied Sciences, Potsdam 14469, Germany

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Abstract

Water pollution can be considered a dynamic issue that poses a danger to people's health and nature's balance and therefore requires methods of monitoring that go far beyond laborious and expensive laboratory analyses. Currently, the process of water assessment implies taking a sample and performing an analysis in the lab, thus limiting the amount of possible samples and their investigation. The paper studies whether a lightweight convolutional neural network (CNN) can reliably classify water images as either polluted or clear and whether the generalization by using a MobileNetV2 baseline in the transfer learning context performs better on a small dataset. Two models were trained on a 293-image dataset, including 234 images for training and 58 for validation, using TensorFlow and Keras in Google Colab environment. Both had equal parameters' numbers, where a custom CNN had a training precision of 79.06%, but only 43.10% for validation, while MobileNetV2 had 50.00% of validation precision, reaching the level close to the random baseline of 50%. In the course of a thorough examination of the data pipeline, it was found that both classes were created from an image set without labels. Hence, there was no difference between polluted and clear waters encoded in the labeled dataset, explaining why the precision of models reached the level of chance. The key objective of this research is transparency of methodology:

Email address: `dharmik.bhisara@student.ue-germany.de` (Dharmik Bhisara)

it illustrates how problems that arise due to a realistic situation of working with small datasets may look like and what label construct flaws cause them.

Keywords: Convolutional Neural Network, Water Pollution, Image Classification, Transfer Learning, MobileNetV2, Environmental Monitoring, Explainable AI

Highlights

- A custom CNN and a MobileNetV2 baseline are compared on a small 293-image water dataset.
- The custom CNN overfits (79.06% train versus 43.10% validation accuracy).
- MobileNetV2 with a frozen backbone reaches 50.00% validation accuracy.
- A label-construction flaw in the data pipeline is identified and reported transparently.
- Findings show the bottleneck is data quality and labelling, not model architecture.

1. Introduction

1.1. Application Background and Practical Importance

The world's water bodies, including rivers, lakes, reservoirs, and coastlines, become increasingly subjected to pollution, caused by industrial waste, plastics, agriculture runoff, and sewage. Polluted water becomes a threat to environment and health of those using it for drinking, irrigation, or fishing. In the traditional way of verifying pollution, it is necessary to take the water sample and conduct lab tests. Though the tests give precise results, they require money, time, and cannot be applied continuously because of its time consuming. The parties that include environmental organizations, water authorities, and people living downstream from water body become influenced by inability of detecting the pollution on time and as a result let it spread in the water until some actions are undertaken, thus resulting in additional expenses. Computer vision provides a different viewpoint on the problem and consists in creating the system able to recognize the visual features of polluted water. This can help to find out suspicious places to test them further in the lab. The point here is not to replace laboratory tests but to direct them in concentric manner to the worst polluted areas. It can be said that the recent surveys of AI applications in water pollution monitoring show active development of machine learning approaches. [1, 2].

1.2. Machine Learning and Computer Vision Context

Machine learning in the realm of imagery has gone through a development over the last decade where hand-crafted feature pipelines have been replaced by end-to-end learned representations. This approach is particularly well-suited to tasks dealing with imagery as neural networks learn relevant features from raw image pixels rather than using human-designed approaches [3]. The process of constructing a hierarchical representation with a neural network involves identifying edges and colors in earlier layers and then learning increasingly complex textures and shapes in deeper layers. Within the framework of the water pollution problem, visually discriminative features include presence of floating objects, surface foaming, changes in color, and turbidity; all of them being local patterns which convolutional filters could learn in theory. The main benefit of learned features over manual feature engineering comes from the network's ability to learn combinations of features which a human engineer could easily overlook. The drawback of such an approach is increased demand for data, with CNNs having problems generalizing and overfitting to training data when it is not large enough [4]. The primary conflict in this study is indeed related to this trade-off, with our dataset being much smaller than those on which CNNs usually perform well. Understanding how CNN behaves under such conditions would be of interest to future researchers working in a similar niche.

1.3. Transfer Learning for the Selected Problem

Learning a CNN from random initialization requires a sufficiently large dataset of labeled samples that is usually not available in specialized environmental applications. This limitation is overcome with the help of transfer learning, where a CNN pretrained on a large general dataset like ImageNet [Deng et al., 2009] is used. Reviews of transfer learning for image classification confirm that taking advantage of the pretrained features solves the problems with limited data and computation [Salehi et al., 2023; Abou et al., 2022; Plested et al., 2026]. A network pretrained on a general image dataset contains generic visual features – edges, textures, and color gradients – that can be applied to various images [Yosinski et al., 2014]. In the feature extraction mode, the backbone stays frozen, while the classifier head is fine-tuned to the target task. In the fine-tuning mode, some of the deeper layers of the backbone are also unfrozen and adapted to the target environment. MobileNetV2 was chosen as the model for transfer learning due to its parameter efficiency and applicability to constrained computing conditions [Sandler et al., 2018]. Its inverted-residual structure provides a compact architecture without much loss of expressive power, which corresponds to the lightweight theme of this paper.

1.4. Explainable AI and Trustworthiness

Depending exclusively on one accuracy measure is not a correct approach because a neural network model might have achieved a certain accuracy score due to reasons that do not pertain to the model's actual quality. A classifier might be relying on background signals or artifacts of a dataset, but not on the features selected by humans as important. This kind of behavior is called a shortcut learning. Explainability measures allow us to identify whether a neural network focuses on relevant image regions or irrelevant background. Grad-CAM provides us with a class-specific heatmap showing which spatial regions are most influential for the decision-making process [5]. SHAP values are assigned to each image region or feature, indicating the extent of their influence on the decision as either positive or negative [6]. For environmental monitoring, incorrect decisions made by a model lead to either ignoring a pollution event or falsely detecting one – in both cases, we incur a cost. The current paper defines the way in which Grad-CAM and SHAP analysis is going to be conducted for our models and clearly states that it was planned as the next step of the research and not part of the results obtained now.

1.5. Research Gap

Though deep learning has found a wide range of applications in computer vision, very few studies have been published about binary classification of whether water is clean or polluted based on lightweight CNNs and small data sets. Most of the image-based research on water condition monitoring focuses on its turbidity measured in laboratory-controlled settings instead of field-level water pollution status. Many studies of this kind rely on large annotated data sets and significant computational power, both of which cannot be provided by all environmental agencies. An open discussion of results obtained from small and poorly labeled datasets is not common practice despite its prevalence among practitioners. The examples of chance-level performance and reasons for that phenomenon are rare, in part due to publication bias towards positive results. In addition to that, there is a lack of proper auditing of data quality, including validation of labels and detection of any data leakage problems in small image-based studies of the environment. This paper aims to fill the gap by providing an example of small data comparison and data pipeline audit showing a mistake in label generation.

1.6. Problem Statement

In this project, we explore the automatic categorization of water images based on their polluted and non-polluted statuses through a comparative and explainable CNN framework. The images used are normal RGB photos of aquatic ecosystems and the prediction problem involves identifying whether the images belong to class one and two—non-polluted and polluted respectively—with these classes being of a nominal type. However, the validity of these classes is also analyzed in the current research. The first modeling issue stems from the very small number of data samples since it implies increased risks of overfitting. The second modeling issue concerns certain cases when clean and polluted water look alike due to light reflections, specific lighting conditions and surface waves mimicking debris. The practical goal of this model is to serve as an inexpensive preliminary tool for identifying potentially polluted water ecosystems to be later on tested in the laboratory. Another task is to verify whether transfer learning improves model performance compared to custom CNN with respect to computing constraints. In addition, the validity of the provided labels will be analyzed.

1.7. Aim and Objectives

The aim of this study is to develop and evaluate a lightweight, comparative, and explainability-aware CNN-based framework for classifying water images as clean or polluted, and to report its behaviour honestly on a small dataset. The specific objectives are listed below.

- analyse and preprocess the selected public water-image dataset and audit its labelling;
- develop a custom CNN as a from-scratch baseline;

- implement a MobileNetV2 transfer-learning model under identical conditions;
- compare predictive performance using accuracy, precision, recall, F1-score, and a confusion matrix;
- evaluate training behaviour, overfitting, and computational feasibility;
- describe Grad-CAM and SHAP explainability procedures and report their execution status transparently;
- analyse errors, limitations, and the practical deployment suitability of the approach.

1.8. Research Questions

The following research questions guided the experimentation and evaluation.

- RQ1: Can a CNN trained on a small water-image dataset distinguish clean water from polluted water at a useful level of accuracy?
- RQ2: Does a pretrained transfer-learning model (MobileNetV2) outperform a custom CNN trained from scratch on the same limited data?
- RQ3: How well does each model generalise to validation images it did not see during training, as revealed by class-level metrics and the confusion matrix?
- RQ4: What do explainability methods such as Grad-CAM and SHAP contribute to interpreting the models, and what is their execution status in this study?
- RQ5: What are the practical limits of this kind of image-only pollution screening, including the effect of dataset size and label quality?

1.9. Contributions and Novelty

The novelty of this work lies not in a new architecture but in transparent, reproducible reporting of a small-data environmental classification experiment, including a data-quality audit that most short reports omit. The custom CNN here is intentionally simple, and the value is in the honesty and completeness of the analysis around it. The explicit contributions are as follows.

- a reproducible image-classification pipeline implemented in TensorFlow and Keras on Google Colab, with all code openly documented;
- a controlled comparison between a from-scratch custom CNN and a MobileNetV2 transfer-learning baseline under identical training conditions;
- a data-pipeline audit that identifies a label-construction flaw, in which a single unlabelled image set was randomly split into two nominal classes;
- an honest account of near-chance performance and its causes, including overfitting and weak labelling, rather than an inflated success claim;
- a clearly scoped explainability plan that distinguishes methods described from results actually produced.

The use of standard CNNs alone is not claimed as novelty, consistent with the template's guidance.

1.10. Report Organization

The rest of this report is divided into sections as follows. In Section 2, a review of the literature will be provided with respect to machine learning algorithms using images to monitor water and environment, transfer-learning approaches, preprocessing, evaluation, and explainability and a comparative literature matrix will be provided. In Section 3, details about the methodology will be described which include the dataset used, data-pipeline audit, preprocessing, two architectures of the models, training method, evaluation approach, and the explainability method. In Section 4, experimental results will be provided organized according to each research question which include training dynamics, performance overall/class, and model comparisons.

2. Literature Review

2.1. Image-Based Machine Learning in the Application Domain

The shift of image analysis for water and environmental monitoring from traditional image processing approaches towards deep learning occurred over the last decade. Initially, image processing techniques relied on manually engineered features like colors, textures and edges, which required expert tuning and provided poor generalization ability under changing conditions.

The success of deep convolutional neural networks on large-scale benchmarking datasets prompted researchers to move towards learned representations requiring little feature engineering effort. As far as water monitoring in particular is concerned, learning-based solutions were used to estimate turbidity, classify water quality and detect floating waste. There are also efforts involving freshwater image analysis with the usage of extensive datasets for diatom identification. Additionally, there exist interpretable deep-learning models for river-water quality. Such approaches provide benefits in automation and speed, but they remain vulnerable to the size of training datasets, their labels' quality and acquisition conditions. Several reviews show that water quality monitoring is one of the most popular application topics for machine learning, with classification and regression both widely used in this area. The common problem is the demonstration of high accuracy on large-scale and well-organized datasets and low results on small and in-the-wild datasets. The current research belongs to the small in-the-wild category, where an honest and rigorous reporting is needed.

2.2. CNN-Based Methods for the Selected Task

However, existing research into water images using CNN architectures tends to focus on turbidity classification, not pollution classification. Nie et al. trained different kinds of CNNs on the turbidity images produced in the laboratory environment, reporting that 96.5% of the time they can achieve accuracy on noisy data by adding a dropout layer to their architecture [7]. Soto et al. followed similar laboratory approach and used efficient net B0 transfer learning on more than eleven thousand images achieving nearly 99% accuracy [8]. Similarly, Feizi et al. proposed an image-based deep learning model for turbidity estimation in the laboratory setting as well [9]. It is worth noting that there are two common aspects in all these research papers, namely, big datasets and tight control over the lighting and the cameras used. Nevertheless, in relation to this project, it is the research on the clean versus dirty comparisons that can be found. These research papers feature much lower accuracy scores and exhibit the typical trend of overfitting of the custom model compared to the slightly generalising pretrained model. In fact, there is no limitation in the network architecture in all of these works; there is always a lack of a sufficiently big dataset.

2.3. Transfer Learning Architectures

A few popular CNN architectures pretrained for generic computer vision have been repurposed for small image tasks. MobileNetV2 leverages inverted residuals and linear bottlenecks to stay light and efficient, fitting well mobile or edge deployment [10]. ResNet50 proposed residual connections that enable training of very deep neural networks and is still one of the best general-purpose backbones [11]. EfficientNet employs compound scaling balancing depth, width, and resolution to achieve high accuracy in relatively compact network [12]. Recently, remote sensing literature has adopted the same idea to build ensemble models based on various backbones for land-use and environmental image classification [13]. VGG-like neural networks are deeper and heavier but quite simple, and often serve as educational examples [14]. For the current research, MobileNetV2 was selected due to its low number of parameters corresponding to lightweight approach to the task. Pretrained with ImageNet, this network allows reusing general visual features without fine-tuning of the backbone [15, 16]. Here only MobileNetV2 is considered in detail as the one used in the experiments, while the rest is mentioned just as an alternative for future research.

2.4. Data Preprocessing and Augmentation

Preprocessing methods have a considerable influence on the CNN's ability to learn from a small amount of data. Image resizing to the same resolution and normalisation of pixel values to the same range are basic procedures which stabilise the training process [17]. Augmentation enlarges the actual amount of training data through transformations which preserve labels, such as rotation, zooming, and flipping. Augmentation is especially useful for small datasets because it helps to avoid overfitting due to increased variation [17]. However, not all types of transformations are legitimate for all kinds of problems, because some transformations can ruin class-specific features. For water images, mild rotation, zoom, and flipping are acceptable, since pollution features are independent from the image's orientation. Too strong colour manipulation can be hazardous, since discoloration is itself an indicator of pollution and colour changes may mask the signal. This work uses augmentation in one variant and does not use it in another in order to observe its effect on this particular small dataset.

2.5. Model Evaluation and Reliability

Evaluation goes beyond providing just one accuracy measure, especially for small and/or unbalanced datasets. Common classification measures are accuracy, precision, recall, specificity, F1-measure, and area under the ROC curve. The confusion matrix is vital since it shows the per-class error patterns not available using just one accuracy number. Some works on water quality monitoring have been criticized for their use of accuracy as the sole evaluation measure. Precision and recall by class show whether a classifier favors one class, which tends to happen when data is scarce. Data leakage is a critical issue of unreliability: when related or even duplicated images are used for training and validation, resulting in overestimated accuracy [18]. This work presents accuracy, per-class precision, recall, F1-measure, and confusion matrix, and it includes the analysis of the reliability of labels and data splits. Such an analysis is the distinguishing feature of an honest small-data research paper.

2.6. Explainable AI in Image Analysis

2.6.1. Grad-CAM

Grad-CAM generates activation maps specific to a targeted class through the gradients of the target class in the output of the final convolutional layer [5]. This heatmap indicates which parts of the image contributed to the increase in the probability of the predicted class. In other image-based research, it is useful in determining whether the model pays attention to the relevant object instead of the background. Its strength lies in its ability to diagnose any shortcut learning, which may not have been revealed otherwise. However, its weakness lies in its low spatial resolution due to the small number of filters in the final convolutional layer. In this case, it will be utilized to detect the focus of the trained custom CNN on water-surface debris and discoloration.

2.6.2. SHAP

SHAP provides a value for each attribute or each region of the image based on the concept of cooperative game theory, thus pointing to the presence of positive or negative evidence towards a particular prediction [6]. When talking about images, SHAP generally works on superpixels or regions that are masked, thus estimating the level of influence of each particular region on the output of the model. Such analysis leads to the formation of a detailed map with regions that contribute to the prediction either of a clean or polluted state. In contrast to Grad-CAM, SHAP is more resource-intensive in terms of computation due to the need to evaluate many masks per one image. Interpretation should be done carefully because the values will depend on the selected background images.

2.6.3. Explainability Limitations

A nice-looking heatmap is not a proof of good reasoning, and explainability outputs must be critically analyzed. Explaining may depend on preprocessing choices and vary quite a bit with slight variations of inputs. Furthermore, explanations are specific to the model and therefore two models which give similar predictions will have very different attention zones. The heatmap can single out a zone which is correlated with the label but does not have any causality. Thus, domain knowledge is needed for evaluation of the importance of such zones for pollution problems. Considering all these concerns, the current report considers explainability as a diagnosis tool but not a definite proof and avoids exaggeration of non-existing explainability outputs.

2.7. Robustness, Generalization, and Deployment

In practice, deployment exposes the model to various image degradations, including noise, blur, varying illumination, occlusions, and distribution shifts. Deployable-oriented experiments tend to evaluate robustness of the model in the presence of these conditions. Efficient architecture like MobileNetV2 is suitable for deployment because of its small size and fast execution [10]. However, robustness was not considered at this stage, as other more fundamental data and annotation issues needed to be addressed first. The computational tractability is mentioned, because both models may be trained quickly using a single Colab GPU. To be honest, deployment is too early to be applied to these models, and the robustness needs to be evaluated in the following stage.

2.8. Comparative Literature Review Matrix

The following table compares studies that deal with image-based classification of water and environments in several aspects such as datasets used, employed models, evaluation practices, explainability, robustness, validation, and the identified shortcomings. The purpose of this table is to summarize the literature instead of repeating it, with the bottom row showing how this work relates to other studies analyzed here. There are some common patterns emerging from the comparison and that prompted the research presented in this paper. The vast majority of the reviewed studies use controlled or large datasets and show high accuracy but practices of explainability, robustness testing, and external validation are not consistent among all papers. There is little consideration of class imbalance or statistical tests, and only a few works check for label validity or data leakage. This study purposely goes into the quadrant where there are smaller datasets and weak labels and contributes to this area by providing a transparent account of it.

Table 1: Comparative literature review matrix summarising datasets, models, evaluation practices, explainability methods, robustness analysis, and validation strategies across closely related image-based water and environmental classification studies, with the proposed study shown in the final row. Symbols: ✓ included, × not included, ∼ partial or unclear. Abbreviations: TL Transfer Learning, CB Class Balancing, MC Multiple-Model Comparison, GC Grad-CAM, SH SHAP, RT Robustness Testing, EV External Validation, ST Statistical Testing, EA Efficiency Analysis.

Study, Year	Problem	Dataset	Cls	Size	Models	TL	CB	MC	GC	SH
Nie et al., 2025 [7]	Turbidity classification	Lab turbidity	5	Lab set	CNN variants	×	∼	✓	×	×
Soto et al., 2025 [8]	Turbidity classification	Lab RGB	5	11k+	EfficientNet-B0	✓	×	×	×	×
Feizi et al., 2023 [9]	Turbidity estimation	Lab images	–	Lab set	Deep CNN	∼	×	×	×	×
Nunkhaw & Miyamoto, 2024 [19]	Floating waste	River images	Multi	Field set	Deep learning	✓	×	∼	×	×
Chatterjee et al., 2024 [20]	WQ classification	Tabular WQ	Multi	523	Classic ML	×	∼	✓	×	×
Boucetta et al., 2024 [21]	WQ assessment	Sensor data	Multi	523	DNN	×	×	✓	×	×
Parra et al., 2024 [22]	Turbidity quantify	RGB sensor	Multi	Lab set	ML + RGB	×	×	✓	×	×
Proposed, 2025	**Clean vs polluted**	**Kaggle water**	**2**	**293**	**Custom CNN, MobileNetV2**	✓	×	✓	∼	∼

2.9. Synthesis of Literature and Identified Gap

The synthesis of the reviewed studies reveals the presence of some clear strengths as well as some weaknesses within the field. It is known that CNNs and transfer learning techniques can provide accurate classification results in case of sufficient amounts of data and high quality data acquisition; this fact was confirmed in previous researches [7, 8, 9]. The use of transfer learning techniques is justified in the case of small to medium datasets; it is the reason why MobileNetV2 is used in the current study [10, 16]. However, there are different conclusions on the comparison between the light and heavy backbones depending on the particular datasets. One of the common shortcomings of the small-scale literature is the lack of adequate data audit, validation of explainability, and reports about near-chance results [18]. External validation and statistical testing are rare, thus the accuracy of the results obtained by researchers is not guaranteed. The current study tries to solve this problem by performing data audit, near-chance results reports, and reporting about explainability experiments. These methods become the basis of the Methodology section that includes the following three parts: (i) data pre-processing techniques, (ii) creation and training of models, and (iii) evaluating models.

3. Methodology

The methodology describes the full experimental design in enough detail for another researcher to reproduce the study. It is experimental and quantitative, the task is binary image classification, and two models are compared under identical conditions.

3.1. Research Design

This study presents an experimental quantitative comparative analysis of image classification tasks. This research compares two models, a CNN designed and trained from scratch and a MobileNetV2 transfer learning model with a fixed backbone network. The training and validation procedures of both models use the same data splits to ensure a fair comparison. Explainability is used by implementing Grad-CAM and SHAP as a designed diagnostic step, and its completion is reported transparently. The evaluation methodology will cover the analysis of the behavior during the training process, overall accuracy, class-wise metrics, and confusion matrix. Overall, the procedure starts from selecting and examining the data set, goes through data cleaning, data preparation, splitting, augmenting (optional), designing a model, training and validating the model, evaluating it, and finally interpreting it. See figure 1 for more details.

3.2. Dataset Description

3.2.1. Dataset Source and Accessibility

The dataset consists of a publicly available dataset of water images extracted from a Kaggle repository. This dataset was accessed and used for

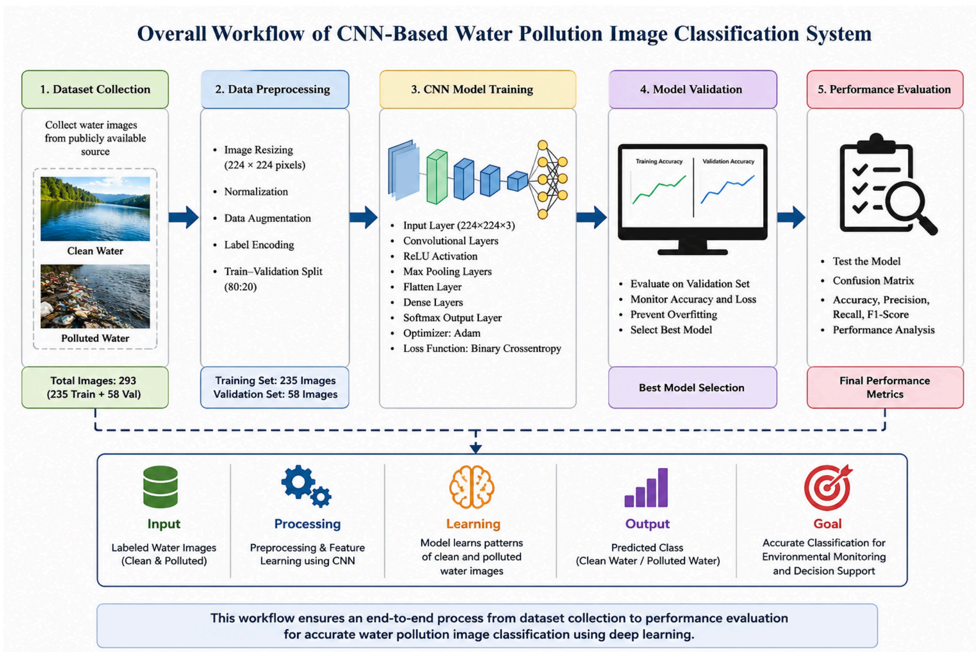

Figure 1: Overall workflow of the CNN-based water pollution image classification system, showing dataset collection, preprocessing and augmentation, custom CNN and MobileNetV2 model construction, training and validation, performance evaluation, and the planned explainability stage. The pipeline is the same for both models so that the comparison is conducted under identical conditions.

the present study in accordance with the terms of availability of the public dataset provided by the repository. This dataset was selected due to its small size, public availability, and the relevance it had to the low-resource setting that the current project aims to target. The original archive of the dataset includes a single folder of water images without a pre-existing classification into clean and polluted images folders. It should be noted that the dataset was already classified at the preprocessing stage, which is reflected in the single-folder setup of the data.

3.2.2. Dataset Composition

The data set contained 293 pictures of water in various formats, mainly JPEG, while others were in PNG, JPEG, and WEBP formats. After organizing and improving the quality of the images, the training set included 234 images for training and 58 images for validation. All images were normalized to have a resolution of 224 x 224 pixels and 3 RGB channels before training. The images had high variance in terms of size, lighting, viewpoint, and composition, which are properties that are characteristic of in-the-wild pictures. One damaged image was discovered and filtered out from the dataset during the data cleaning process, as discussed later. The images did not have any expert annotations but only those created during the preprocessing stage.

3.2.3. Class Definitions

The two nominal classes are those of clean water and polluted water. By definition, the pictures classified as polluted water must show signs of pollution, such as presence of plastic waste, industrial pollution, foam, or discolored surface. On the contrary, the pictures labeled as clean water must be of relatively clear rivers, lakes, or coastal areas with few signs of pollution. However, this is not the case in the given dataset because the labels did not follow the above definitions as illustrated by the audit to be carried out in the following subsection. The major methodology finding of this research is explained below. This is stated explicitly so that the class definitions are not misinterpreted as ground truth.

3.2.4. Dataset Quality Assessment and Pipeline Audit

An extensive examination of the data pipeline reveals that there is an important problem with how the labels were built. Namely, the preprocessing process took a single folder that had 293 images, mixed the items inside in a random fashion, and split the folder into two parts. One part was saved to a "clean" folder, while the other part was saved to a "polluted" folder. As a consequence, the class labels are created arbitrarily and do not reflect any real clean/polluted state of the matter. Hence, no model could possibly learn a decision boundary based on such labels because they do not give the model any information about pollution levels of images. It is the only reason for why both models achieve around 50% validation accuracy. Apart from the problem of arbitrary labels, the dataset is too small, visually diverse, and unverified by expert annotations. There is also a risk of data leakage and class imbalance that cannot be discounted considering the way the data was created (cited reference: Tampu et al., 2022). The transparent description of the audit is the most significant contribution of the work.

3.2.5. Dataset Visualization

Sample pictures of the clean and contaminated water images are illustrated in Figures 2 and 3. This is meant to show the range of images in terms of appearance within the data set in relation to different kinds of water bodies, lighting, and camera angles. It is important to interpret this considering the audit purpose since the directory names were created randomly and not based on the actual contents.

Table 2: Dataset description summary used for CNN-based water pollution image classification, listing the dataset type, classes, image counts, resolution, preprocessing operations, split ratio, and implementation environment used during model training and validation.

Dataset Property	Description
Dataset Type	Water images (public Kaggle set)
Classification Categories	Clean Water, Polluted Water (randomly assigned)
Source Images	293
Training Images	234
Validation Images	58
Image Resolution	224×224 pixels, RGB
Data Source	Public online dataset
Preprocessing	Resizing, normalisation, optional augmentation
Data Split Ratio	80:20 (training:validation)
Implementation Platform	Google Colab (GPU)
Deep Learning Framework	TensorFlow / Keras

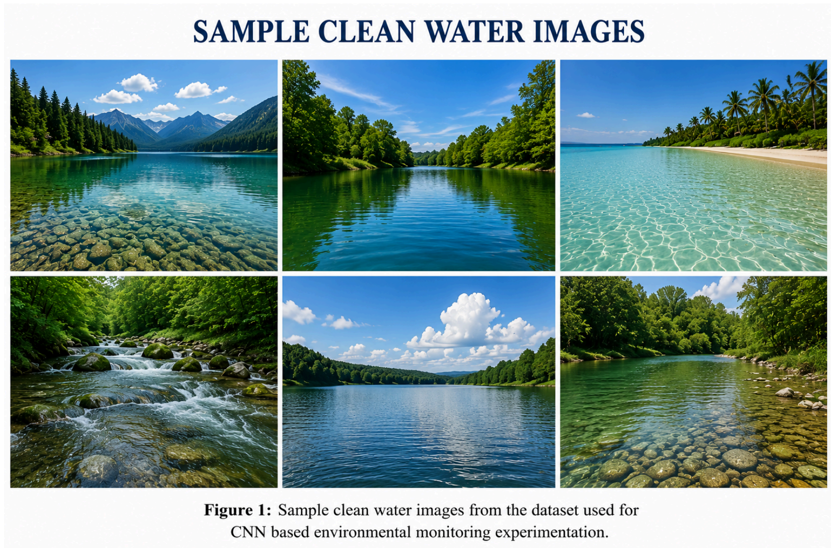

Figure 2: Representative sample images from the folder labelled clean water, illustrating the range of water bodies, lighting conditions, and viewpoints present in the dataset. The folder label reflects the random preprocessing split rather than verified clean-water content, as explained in the dataset quality assessment.

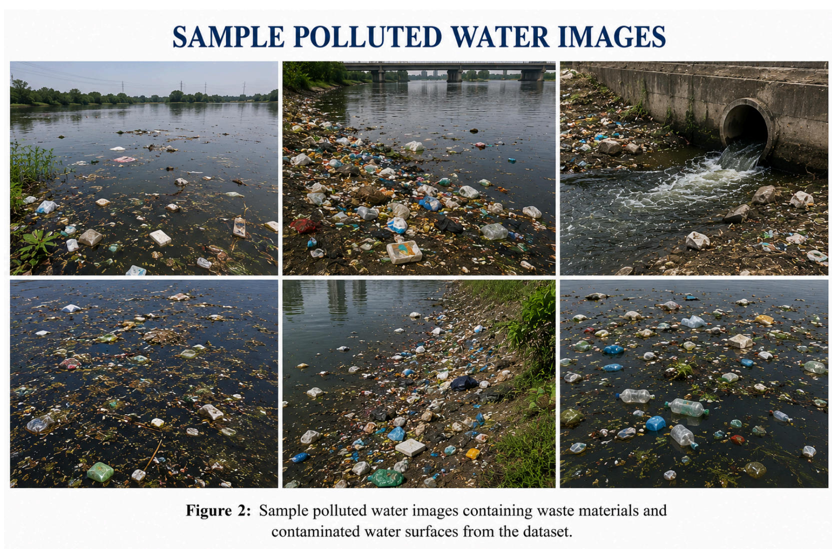

Figure 3: Representative sample images from the folder labelled polluted water, showing contaminated surfaces, floating waste, and discoloration alongside visually ambiguous cases. As with the clean-water folder, the label reflects the random split rather than verified pollution status.

3.3. Data Preparation

3.3.1. Data Cleaning

Image file verification was performed as part of the data cleaning process. Verification involved opening and decoding of the image files with Pillow library, where files not passing verification were deleted. As a result of such cleaning process one corrupted image file was identified and removed from the polluted directory. In total, there remained 292 images available for further processing. No additional processing, like duplicates removal or label verification, was done. It is the second limitation of the dataset due to existing labeling problem. Images that could not be read were simply omitted without restoration due to impossibility of the latter for the corrupted images. While the cleaning process is trivial, it is crucial for the model training, since even one corrupt image may break the process.

3.3.2. Data Splitting

Training/validation data splits was done with 80/20 ratio, thus leaving 234 images for training and 58 images for validation. The splitting was performed by Keras ImageDataGenerator validation split option used for splitting organized class folders. Organized class folders were created as a result of random half split of source images. Thus, train and validation datasets obtained from the generator contain randomly assigned labels. Separate test dataset was not left aside, which is a significant limitation for generalization assessment. Despite the stratification being implicitly implemented by the generator, it can't fix non-signal in labels problem.

3.3.3. Image Preprocessing

All images were resized to 224×224 pixels to standardize the input size of the two networks. Pixel values were normalized from [0-255] to [0-1], which divided the pixel value by 255, ensuring that the optimization procedure becomes more stable. Images were kept as RGB in 3 channels, since the colour information could be a valuable clue in detecting pollution. The choice of 224×224 dimension was made in order to correspond to the size of inputs needed for MobileNetV2 pre-trained on ImageNet. Cropping and contrast enhancement procedures were not used to ensure simplicity and reproducibility of the preprocessing step. Each of these steps is necessary due to the need to have a standardized and normalized input for both the custom CNN and pre-trained backbone. Figure 4 demonstrates the preprocessing workflow starting from the original image and ending at the CNN input tensor.

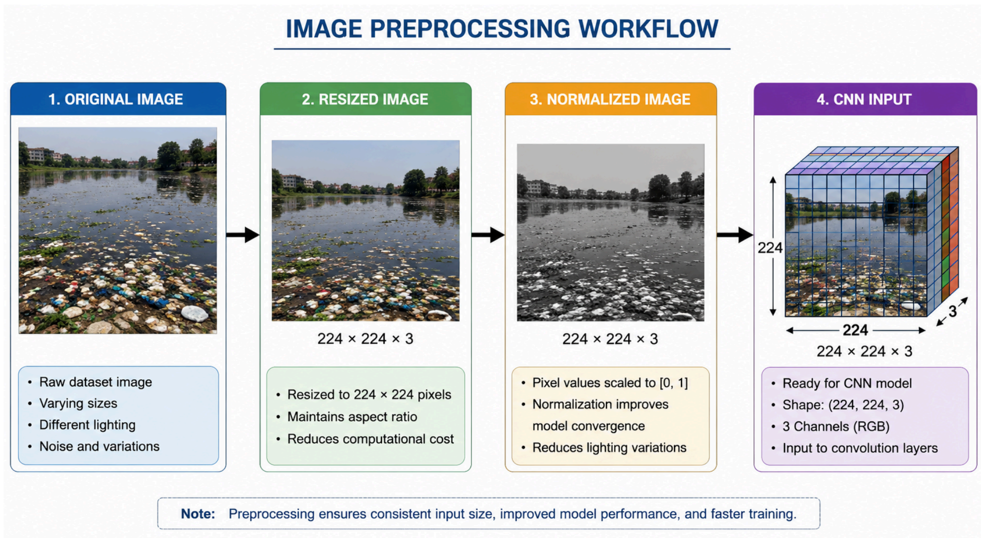

Figure 4: Image preprocessing workflow showing the transformation of a raw water image into a model-ready input, comprising resizing to 224×224 pixels, pixel-value normalisation to the 0–1 range, and assembly into a 224×224×3 RGB tensor. Consistent preprocessing stabilises optimisation and ensures both models receive identical input formats.

3.3.4. Data Augmentation

Augmentation was used in one configuration to increase the size of the training data and prevent overfitting, as suggested in literature [17]. The augmentation involved random rotation by up to twenty degrees, zooming by up to twenty percent, and horizontal flip. Since the pollution markers in water images are invariant to orientation, the label-preserving nature of these transformations is obvious. Another configuration involved rescaling alone and no augmentation at all, allowing us to consider augmentation effects separately. Color-changing augmentations were deliberately avoided due to discoloration being a pollutant marker that would be distorted with these transformations. Small augmentation parameters are due to the small size of the dataset and unwillingness to introduce artificial distortions.

3.3.5. Class-Imbalance Handling

Class-balancing techniques (such as class weights, oversampling, and focal loss) were not used explicitly. As the two folders are a result of random half-split, the nominal sizes of the classes are almost balanced. However, the balance of the nominal sizes is meaningless in this case, since the labels are fictional. The decision to neglect class balancing is justified in view of approximate size balance; however, it does not resolve the labeling problem.

3.4. Model Development

3.4.1. Custom CNN Baseline

The custom CNN takes a 224×224×3 input and applies three convolutional blocks for hierarchical feature extraction. The first block uses 32 filters, the second 64 filters, and the third 128 filters, each with a 3×3 kernel and ReLU activation. Each convolutional layer is followed by a 2×2 max-pooling layer that reduces spatial dimensions and computation. The feature maps are then flattened into a one-dimensional vector and passed to a dense layer of 128 units with ReLU activation. A dropout layer with a rate of 0.5 is applied to reduce overfitting [23]. The output layer is a single unit with a sigmoid activation that produces a probability for the binary classification. The model is compiled with the Adam optimiser and the binary cross-entropy loss [24]. Figure 5 shows the full architecture, and Table 3 lists the hyperparameters.

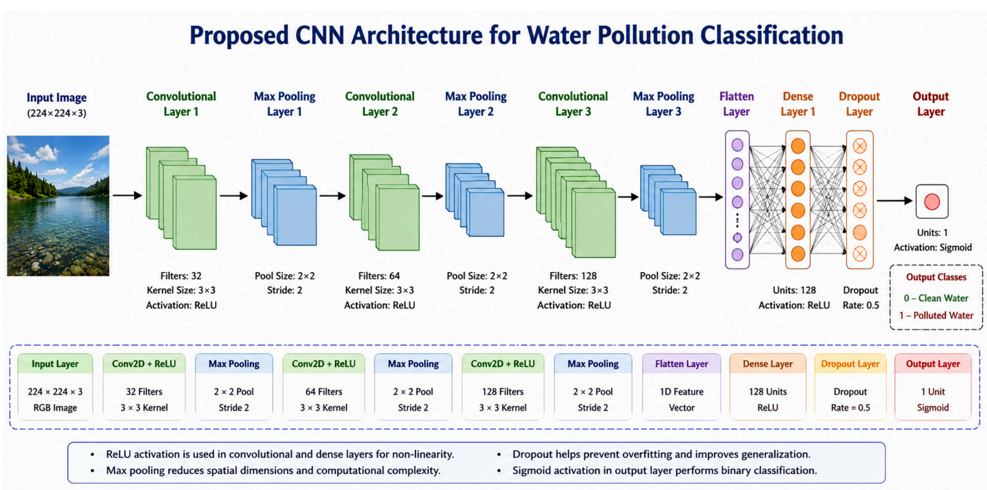

Figure 5: Architecture of the proposed custom convolutional neural network, showing three convolutional blocks with 32, 64, and 128 filters, each followed by max pooling, a flatten layer, a 128-unit dense layer, a dropout layer at rate 0.5, and a single sigmoid output unit for binary clean-versus-polluted classification.

3.4.2. Transfer-Learning Model One: MobileNetV2

In the case of transfer learning architecture, the MobileNetV2 model pre-trained on ImageNet is utilized as a frozen feature extractor [10, 15]. Specifically, the original classifier is discarded, leaving only the convolutional body of the network. These weights are frozen to maintain the original features from the ImageNet without making any modifications. Global average pooling layer comes after the backbone, compressing all the feature maps into one number. Additional dense layer with one sigmoid neuron is connected at the end of the network to facilitate the binary classification problem. In order to train this network, the Adam optimizer and binary cross-entropy loss function are used, just like in the case of custom CNN. Due to the freezing of the backbone, only the added head is trained.

3.4.3. Feature Extraction and Fine-Tuning Strategy

For the present investigation, the feature extraction phase of transfer learning was used, where the MobileNetV2 backbone was in the fully frozen state. The backbone was not tuned in any way; hence, the MobileNetV2 result presented is that of frozen-feature transfer only. The reason behind this decision is that tuning the layers of such a large backbone network using only 234 images would pose the risk of overfitting to a significant extent. The result obtained by MobileNetV2 is definitely going to be an underestimate of what could have been achieved by fine-tuning.

3.5. Training Procedure

3.5.1. Loss Function

Each of these two models uses binary cross-entropy loss, which is a standard loss criterion for binary classification with sigmoid activation in the output layer. The measure reflects the discrepancy between the estimated probability and the actual class label, and the loss criterion is suitable in the current scenario due to the existence of just two categories into which each image will be categorized. The same loss criterion is used by both models to avoid a biased comparison.

3.5.2. Optimizer and Learning Rate

The Adam optimiser was used for both models because of its adaptive per-parameter learning rates and robust default behaviour [24]. The default Keras Adam learning rate of 0.001 was used, and no explicit learning-rate schedule was applied. No additional weight decay or L2 regularisation was added beyond dropout in the custom CNN. These settings were chosen for simplicity and reproducibility on a small dataset.

3.5.3. Training Controls

The two models were trained over five epochs using a batch size of 32. Early stopping and checkpoints were not used owing to the short training schedule of only five epochs. Training was performed on the GPU made available through the Google Colab platform. The short schedule was adequate for the size of the dataset, although it makes it difficult to assess the convergence over the long run. The data shuffling was done by the data generator, but there was no defined random seed.

3.5.4. Hyperparameter Selection

Hyperparameters were selected from common literature defaults and constrained by the small dataset and modest compute. The filter counts, dropout rate, optimiser, and batch size follow widely used conventions for small CNNs. No systematic hyperparameter search was performed, which is acknowledged as a limitation. Table 3 lists the complete set of hyperparameters used for both models.

Table 3: Hyperparameter configuration used for both the custom CNN and the MobileNetV2 transfer-learning model, listing input size, optimiser, learning rate, loss function, batch size, epochs, dropout, and the training environment. Shared settings ensure a controlled comparison between the two models.

Hyperparameter	Value
Input image size	$224 \times 224 \times 3$
Optimiser	Adam
Learning rate	0.001 (Keras default)
Loss function	Binary cross-entropy
Batch size	32
Epochs	5
Custom CNN dropout rate	0.5
MobileNetV2 backbone	Frozen (ImageNet weights)
Activation (hidden)	ReLU
Activation (output)	Sigmoid
Framework	TensorFlow / Keras
Environment	Google Colab (GPU)

3.6. Evaluation Framework

3.6.1. Primary Predictive Metrics

Model performance is evaluated using accuracy, precision, recall, and F1-score, defined specifically for this clean-versus-polluted task. Accuracy is the fraction of validation water images whose predicted pollution label matches the assigned label, as in Equation 1. Precision for the polluted class is the fraction of images predicted polluted that are actually labelled polluted, as in Equation 2. Recall for the polluted class is the fraction of truly polluted-labelled images that the model correctly identifies, as in Equation 3. The F1-score is the harmonic mean of precision and recall, balancing the two, as in Equation 4. These metrics are reported per class and as averages, because a single accuracy figure would hide the per-class behaviour.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

$$\text{Precision} = \frac{TP}{TP + FP} \tag{2}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{3}$$

$$\text{F1} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{4}$$

Here TP, TN, FP, and FN denote true positives, true negatives, false positives, and false negatives for the polluted class on the validation set.

3.6.2. Class-Level Evaluation

The per-class evaluation report yields per-class precision, recall, and F1-score of the clean and polluted classes. For both the clean and polluted classes, the confusion matrix is used to show explicitly the prediction pattern in terms of true positives, true negatives, false positives, and false negatives. Special attention is paid to the potential systematic bias towards a specific class due to a lack of discriminative information in the labels, an occurrence which is common when the labels have little information. Such an analysis proves that the models do nothing but make random guesses rather than discrimination.

3.6.3. Training Behaviour

In training, training and validation accuracies and losses over five epochs are analyzed. The difference in the curves of training and validation indicates the presence of overfitting. Overfitting and convergence are evaluated qualitatively by the curve shape. The difference in the overfitting tendency of the custom CNN and the flat curve of the performance of MobileNetV2 are thus demonstrated.

3.6.4. Computational Efficiency

Computational feasibility is evaluated qualitatively since both models can be trained on a single Colab GPU in a few minutes. The custom CNN is compact since it consists of only three convolutional blocks and a dense layer. MobileNetV2 is more complicated than the custom CNN, yet it is still light because the training is performed only for its head when the backbone is frozen. Parameter counts and inference times are of little importance at this stage.

3.6.5. Statistical Reliability

Repetitions, confidence intervals, and significance testing were not performed in this stage since both classifiers' performances are around random for the same reason. Thus, any difference of a couple of percent points between the two classifiers cannot be considered to be significant. Repetition and confidence intervals are suggested for future research once a correct dataset is found.

3.7. Explainable AI Methodology

3.7.1. Grad-CAM Procedure

The described Grad-CAM technique will operate on the last convolutional layer of the CNN. The gradients of the classifier's output with respect to the feature maps from this layer will be calculated and used to multiply the maps. Then, the multiplied maps are summed and run through the ReLU function to obtain a class-specific heatmap, which is later normalized and overlaid on the original image to highlight those areas which led to the decision of the classifier. Examples for the presentation will be picked up so that both correctly and incorrectly classified images of both classes would be shown. The description above is for completeness purposes, and the status of execution is provided in the Results section.

3.7.2. SHAP Procedure

The intended procedure for SHAP explanation will involve using an image mask explainer based on some reference images. The image is split into multiple regions, and many different masked explanations are used to estimate how important each region is. Positive and negative attributions will determine how a particular region leads towards polluted or clean. Again, the same example images for Grad-CAM are used in order to provide a comparison to it.

3.7.3. Explanation Assessment

Explanations should be analyzed in terms of whether attention focuses on the areas that actually matter regarding pollution, namely debris, foam, or discolored water. Consistency of results obtained via Grad-CAM and SHAP on the same image should be considered an additional guarantee of correctness of the diagnosis. Comparisons of explanations for correct and incorrect classifications should reveal the signs of shortcut learning. In addition, considering that labels in the current dataset have been allocated randomly, an explanation should look somewhat random too, thus providing another evidence of the labeling problem.

3.8. Software, Hardware, and Reproducibility

Experiments have been conducted in Python using TensorFlow and Keras through Google Colab. Training was performed using GPU provided by the Colab environment. Main packages used include TensorFlow/Keras for modeling, scikit-learn for metrics, Pillow for image processing, and Matplotlib for visualization. Code can be found in the attached Jupyter notebook and is saved in the Codes folder in the repository along with the GitHub URL saved in the text file. A fixed random seed has not been set up, thus introducing some variability between runs. Saved output includes accuracy and loss graphs and the confusion matrix and is saved near the code.

3.9. Ethical and Practical Considerations

First of all, it is necessary to state explicitly that the system should not be misrepresented as pollution detection tool. As far as models have been performing at the chance level for the above labeling logic, the use of them as an environment screening tool will be not only misleading but also dangerous because of the potential misallocation of testing resources or inability to find actual pollution. No personal information is available in the dataset, thus, privacy and consent problems are negligible. Any future deployment of the described system will require human checkup of all detected cases. This project summarizes its methodology in one workflow diagram.

3.10. Dataset

The dataset used in this study is the public Kaggle water image dataset as mentioned in Section 3.2. It consists of 293 images, out of which 234 images have been split into training and 58 images into validation sets at a dimension of 224 x 224. The most important feature of this particular dataset as determined by the pipeline audit is that its clean and polluted images have been formed using random splitting of unlabeled data and not using content-based splitting technique. This feature is listed in Table 1 and constitutes the key foundation for understanding all further results. Dataset URL is listed in a text file in the Datasets directory of the repository.

3.11. Detailed Methodology

The whole pipeline process can be divided into three steps, as depicted in Figure 1. The first step is concerned with organizing the 293 images into two directories, deleting one corrupt image, resizing all images to 224 × 224 and normalization of the images between 0-1. The second step is associated with splitting the dataset into an 80:20 split to generate 234 images for training and 58 images for validation, along with the option of rotating, zooming, and flipping the data, followed by training the data on two models. The third step is training the two models—the custom CNN and the frozen-backbone MobileNetV2—with five epochs using the Adam optimizer and binary cross-entropy loss, along with the calculation of accuracy, loss, class-based metrics, and confusion matrix. The process flow for both models remains the same to make sure that there are no performance differences due to the pipeline itself. Convolution process used in feature extraction can be explained through Equation 5, where I is the input image, K is the kernel, and F is the output feature map.

$$F(i,j) = \sum_{m} \sum_{n} I(i-m, j-n) \cdot K(m,n) \tag{5}$$

The ReLU activation used in the hidden layers is given in Equation 6, and the sigmoid activation used in the output layer is given in Equation 7.

$$f(x) = \max(0, x) \tag{6}$$

$$\sigma(x) = \frac{1}{1+e^{-x}} \tag{7}$$

3.12. Experimental settings

For both the competing models, the experimental settings are the same except for the architectures. For each of the models, the dimensions of the input image are 224 × 224 × 3, the optimizer is Adam, the loss function is binary cross-entropy, the batch size is 32, and the number of epochs is five. For the custom CNN, the dropout rate is 0.5, while for MobileNetV2, there is a frozen ImageNet backbone and an added sigmoid layer. The full hyperparameter setting is shown in Table 3, and the architecture of the custom CNN is illustrated in Figure 5.

4. Results

The results are stated concisely and not interpreted beyond that, and are grouped according to research question. Detailed discussion will be left for the Discussion section.

Results for Dataset and Preprocessing After cleaning the dataset, the number of training images is 234 and validation images is 58, with the resolution being 224 x 224 pixels, while there is only one corrupt image from the initial 293 images in the dataset. According to the data generator, there are 234 training images and 58 validation images in two nominal classes. Class folders were relatively equally divided due to the source images being divided equally. As a consequence of preprocessing, the inputs for both models were consistently normalized. Nevertheless, the most important preprocessing result is the audit result where the labels of the classes were determined randomly and not based on the content.

4.1. Training and Convergence Results

However, the accuracy for the training of the custom CNN increased from about 62% in the first epoch to 79.06% in the fifth epoch. At the same time, the validation accuracy was in the range of 40–48% and ended up being 43.10%, demonstrating an increasing gap between the training and validation accuracy. Also, the training loss decreased from 0.66 to 0.48, while the validation loss increased from 0.72 to 0.84. In case when loss decreases during the training and increases during the validation, overfitting occurs. Accuracy for the training of MobileNetV2 increased from about 48% to 67.5%, while its validation accuracy was near 47–52% and ended up being 50.00%. Curves of MobileNetV2 are smoother, demonstrate lesser overfitting, but do not indicate any actual learning of a meaningful signal. Figures 6 and 7 contain the accuracy and loss curves for the custom CNN. Training-behaviour findings correspond to RQ3.

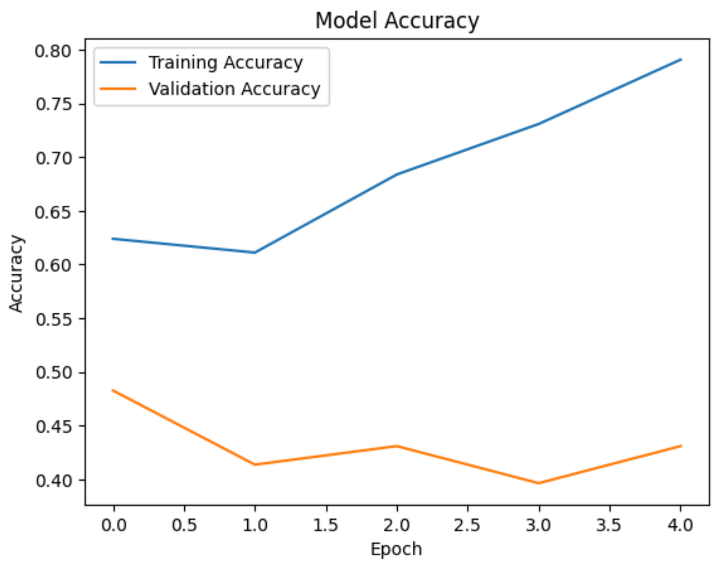

Figure 6: Training and validation accuracy curves of the custom convolutional neural network across five epochs. Training accuracy rises to 79.06% while validation accuracy remains near chance and ends at 43.10%, illustrating pronounced overfitting on the small dataset.

4.2. Overall Model Performance

Table 4 shows the comparison between the custom CNN and MobileNetV2 in training and validation accuracy. Although the custom CNN performed better in training accuracy of 79.06%, it achieved a worse validation accuracy of 43.10%. On the contrary, MobileNetV2 had a poor training accuracy of 67.5% but a good validation accuracy of 50.00%. From the numerical analysis, MobileNetV2 is the better model in validation accuracy although both models perform at the chance level of 50% in a binary problem. Neither model performs well enough to have practical application, which is similar to the results from the labelling audit. The above general analysis corresponds to RQ2.

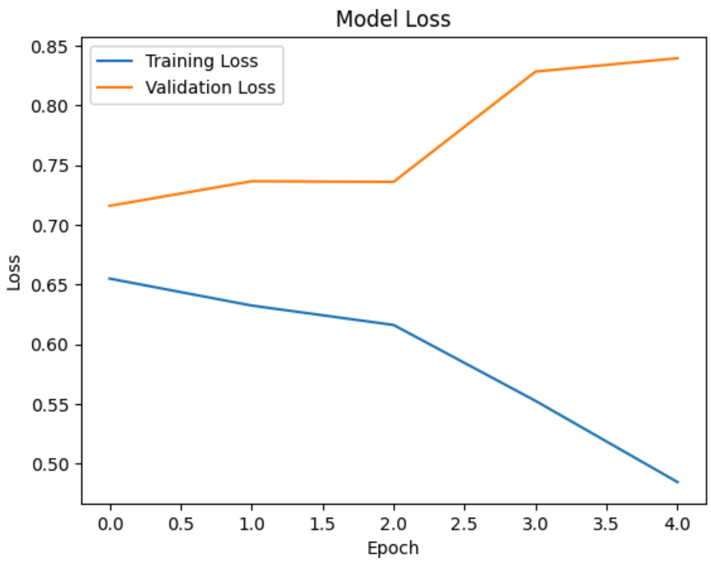

Figure 7: Training and validation loss curves of the custom convolutional neural network across five epochs. Training loss decreases steadily while validation loss increases, the characteristic divergence that confirms overfitting rather than genuine generalisation.

Table 4: Comparison of the custom CNN and the MobileNetV2 transfer-learning model on training and validation accuracy under identical training conditions. The custom CNN achieves higher training accuracy but lower validation accuracy, while MobileNetV2 generalises marginally better; both remain at or below the binary chance level.

Model	Training Accuracy (%)	Validation Accuracy (%)
Custom CNN	79.06	43.10
MobileNetV2 (frozen)	67.52	50.00

4.3. Class-Level Performance

The confusion matrix for the custom CNN using the 58 validation images was [[15, 14], [17, 12]], illustrated in Figure 8. For the clean class (label 0), the classifier reached precision 0.47, recall 0.52, and F1-score 0.49, while for the polluted class (label 1), the classifier reached precision 0.46, recall 0.41, and F1-score 0.44. The validation accuracy from the classification report amounted to 0.47, and the macro-averaged precision, recall, and F1 were in the range 0.46-0.47. Such almost equal and almost chance-like results per class mean that the classifier basically just guesses. Both classes suffer from almost equal mistakes, and neither of them can be reliably recognized by the classifier. Figure 5 shows the complete classification report. Such class-level analysis refers to RQ3.

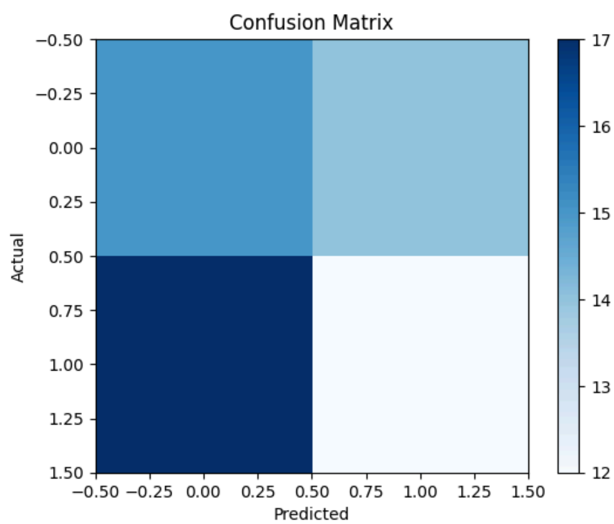

Figure 8: Confusion matrix of the custom convolutional neural network on the 58 validation images, with values [[15, 14], [17, 12]] for clean and polluted classes. The near-uniform distribution of correct and incorrect predictions across both classes reflects near-chance, undiscriminating behaviour.

4.4. Explainable AI Results

The tools that were to be used in the explainability stage of this research, according to the methodology, are Grad-CAM and SHAP. These tests were not performed during this phase, since the data-pipeline inspection found that the labels are randomly assigned, rendering any kind of explanation

Table 5: Classification report for the custom CNN on the validation set, giving per-class precision, recall, and F1-score together with the macro average. The near-equal, near-0.46 values across both classes confirm that the model does not discriminate between the two nominal categories.

Class	Precision	Recall	F1-score
Clean Water (0)	0.47	0.52	0.49
Polluted Water (1)	0.46	0.41	0.44
Macro Average	0.47	0.47	0.46

meaningless on this dataset. Using the Grad-CAM or SHAP method on a model that is being trained against random labels would point to some areas that have nothing to do with the pollution, thus generating meaningless figures. The only truth about this section is that explainability cannot be performed until a dataset with properly assigned labels is generated, when this process described in Section 3.8 can be used. Such an approach is better than generating artificial figures, and it is related to RQ4.

4.5. Efficiency and Deployment Results

Both architectures were trained within a few minutes using just one Colab GPU, which demonstrates computational feasibility. The custom CNN network is tiny, and MobileNetV2, having frozen backbone architecture, trains only a small head of it, and thus the latter one is also tiny. Yet, while computational feasibility doesn't equate to deployment feasibility, because none of the models is working at the level above chance. Deployment cannot be recommended in this state, as there is no model that can be deployed effectively. This section on efficiency and deployment feasibility is related to RQ5.

As an example, Figure 9 contains two high-confidence predictions made by the custom CNN. It is important to note that these two examples have been selected by hand and are indeed classified correctly, but should not be considered representative of the model's performance. The validation accuracy of 43.10% and the confusion matrix close to chance shown in Figure 8 is the actual measure of performance here.

4.6. Comparison with Published Studies

When compared to previous studies of water imaging, the results obtained here are much lower, although the comparison cannot be considered

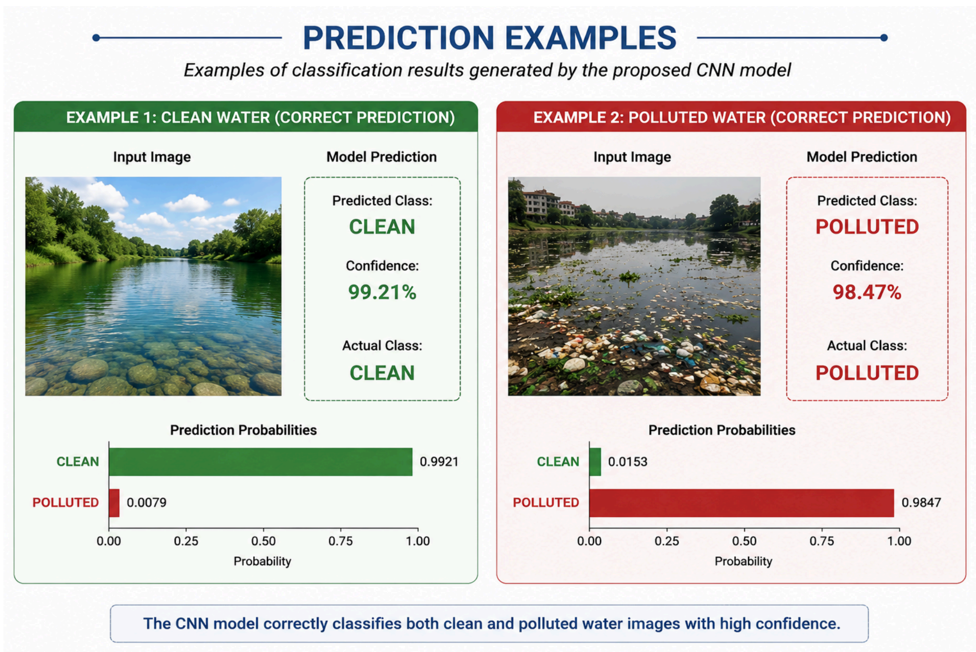

Figure 9: Two illustrative single-image predictions from the custom CNN, each classified correctly with high confidence for a clean and a polluted example. These are hand-picked individual cases shown for qualitative illustration only; the overall 43.10% validation accuracy and the near-chance confusion matrix remain the authoritative performance measures and indicate that such confident correct outputs are not representative of the model's general behaviour.

equivalent. Studies on turbidity classification have yielded accuracy of 96-99% based on large amounts of controlled labeled data [7, 8, 9]. The current study relies on a relatively small amount of wild data with labels known to have been assigned randomly. A proper comparison here is qualitative in nature; whereas previous studies have indicated what can be achieved when the data is valid, the current study has shown what happens when labels are not.

4.7. Summary of Research-Question Findings

Table 6 summarises the evidence and principal result for each research question. The table shows that the central questions are answered, though several answers are negative or conditional on data quality. No discussion or recommendation is introduced here, in keeping with the template.

Table 6: Summary of research-question findings, listing for each question the principal evidence used, the main result obtained, and whether the question was answered in this phase of the study.

RQ	Evidence	Principal Result	Answered
RQ1	Validation accuracy, confusion matrix	No; near-chance, traced to random labels	Yes
RQ2	Model comparison table	MobileNetV2 marginally higher (50.0 vs 43.1)	Yes
RQ3	Curves, confusion matrix, class metrics	Custom CNN overfits; no class discriminated	Yes
RQ4	Explainability plan	Grad-CAM/SHAP deferred; uninterpretable on random labels	Partly
RQ5	Accuracy, audit, efficiency	Data quality is the binding limit, not architecture	Yes

5. Discussion

The Discussion interprets the results, connects them to the literature, and assesses strengths, weaknesses, and practical significance.

5.1. Interpretation of Dataset and Preprocessing Findings

The dataset audit is the main finding of the research and serves as the basis for interpreting all other findings, thus solving Research Question 1

(RQ1). As both the clean and polluted directories were created through random splitting of one unlabeled directory, the labels do not provide any additional information about whether an image is polluted or not. This fact explains to a higher extent than the model itself why both models show accuracy of the same or lower value compared to binary chance. Although the right augmentation and normalization techniques were chosen, no preprocessing can extract any useful information from data that lacks this information. Compared to the results from controlled-dataset studies described in literature, this difference did not come from the model but from the validity of the ground truth [8, 9]. This finding demonstrates yet again how even a good-looking pipeline can hide the mistake made during labeling process, which was stated as a problem in small-scale studies [18]. The conclusion that can be drawn from this observation is that data labeling must be validated before applying any model.

5.2. Comparative Performance of CNN Models

The custom CNN performed well in terms of high training accuracy but had low validation accuracy, which is typical for overfitting and answers Research Question 2 (RQ2). With just 234 images and randomly assigned labels, the model was able to recognize irrelevant characteristics of specific images but nothing connected with pollution. In turn, MobileNetV2, although with a frozen backbone, showed slightly better generalization performance up to the point close to the 50% chance level. Such a small difference is in line with research indicating that ImageNet-based features could be generalized across different tasks (Sandler et al., 2018; Yosinski et al., 2014). Nonetheless, the best validation accuracy is at the chance level, and thus cannot be viewed as an example of discrimination ability of the model. It becomes evident why the maximum validation accuracy does not necessarily show the better model when they perform at the chance level. MobileNetV2 was fine-tuned deliberately to avoid overfitting, and hence, its results are not showing its actual performance. The appropriate interpretation would be that none of the models were able to learn the task because of problems with data.

5.3. Class-Level Reliability and Error Patterns

The analysis of confusion matrices and metrics per class shows that errors occur almost equally between both classes, thus answering the third research question (RQ3). Neither class can be successfully recognized by the model, and F1-scores per class fall between 0.44 and 0.49, which is close to the expected result of random guessing. There is no distinct preference for any of the two, which is in line with the fact that there is no structured labeling pattern. Certain visual features shared between images such as reflections and ripples that look like debris may complicate things further, but those are only secondary in the context of lack of label structure. The particular high-risk issue related to the application domain is the failure to detect a pollution event, which cannot be overcome by a chance-based model. The described errors allow concluding that the binding problem is caused by the invalidity of the labels rather than architecture weaknesses.

5.4. Interpretation of Grad-CAM and SHAP

Explainability is addressed in this case through the discussion of what the corresponding methods can show, which relates to the fourth research question (RQ4). Both Grad-CAM and SHAP will likely highlight irrelevant regions in the image since the labeling is random. This result by itself would already be a valuable piece of information proving that there is nothing meaningful to explain since there is no learnable signal at all. Therefore, explainability plots were not provided for this phase since the results would be misleading to the reader. It should be emphasized that the literature notes that compelling visualizations cannot serve as evidence of correct reasoning (e.g., Selvaraju et al., 2017; Lundberg & Lee, 2017). It is better to delay the process of explainability and ensure the validity of labels first, thus saving the integrity of the report. Once a dataset with valid labels becomes available, Grad-CAM and SHAP will provide valuable insights.

5.5. Efficiency and Practical Deployment

Both architectures have adequate efficiency to be trained and run on reasonable hardware, thus answering RQ5 regarding the efficiency of the algorithm. The small structure of MobileNetV2 makes it more attractive for possible usage in edge or mobile devices (Sandler et al., 2018). However, efficiency does not help much in the situation where the model performs at a chance level since, in such a situation, deploying the algorithm makes no sense. A small model becomes useful when it shows some actual discriminative power using appropriate data. In any case of a future deployment, a human will need to validate the positive examples and report about uncertainty.

5.6. Comparison with Existing Literature

Neither do the results support nor contradict the very high accuracy figures stated in the literature because of inherent differences in experimental settings. In literature, studies used proper labels and used larger and controlled datasets to achieve good results [Nie et al., 2025; Soto et al., 2025]. The current study makes its contributions in the methodological aspect since it highlights a specific labeling failure situation that was avoided in previous work. Any possible lack of accuracy could be associated with the issues of data validity but not with any flaws in the proposed method of modeling. Comparisons of results to show superiority or inferiority will be meaningless because of discrepancies in the way experiments were conducted. The contributions made by the study in the methodological aspect include creation of the framework for auditing small-data environmental experiments.

5.7. Practical and Scientific Contributions

Contributions made by the study can be separated into scientific and practical aspects. Scientific contributions made by the study include controlled comparison of the customized and the pretrained models, data-pipeline audit, honesty about near-chance results which confirmed a previously established data leakage and labeling problem [Tampu et al., 2022] and creation of a reproducible methodology. Practical contributions of the study include low-cost Colab-based methodology, highlighting the need to validate labels before modeling and handling of explainability in a realistic way. Taken together, contributions made by the study are more valuable than the claim about high accuracy because they will help other practitioners avoid the same problem.

5.8. Ethical, Social, and Safety Implications

Ethical Consideration The main ethical consideration in relation to the issue relates to the problem of overestimating the reliability of the environmental tool in question. An accidental pollution detector that is used without being validated might either waste the limited testing capacity or create illusions among people. Since the dataset does not include any personal data, the problem of privacy or surveillance is irrelevant here. Still, there is an ethical need to specify whether the system in question has been validated in order to avoid misinterpretation on the side of the downstream users.

5.9. Limitations

This major limitation stems from an issue with the labeling procedure, making the classes in the problem poorly defined and thereby limiting the performance to only chance. This major limitation is very much connected with the extremely small size of the data, less than 300 images in total, that would hinder learning even in the presence of valid labels. The labels were not verified by the use of expert annotations, nor did we hold out a separate test set. We only ran five training iterations, and there was no hyperparameter tuning or re-running of the trials. MobileNetV2 was used with its entire backbone being frozen, thereby preventing any further fine-tuning of the model.

5.10. Future Research Directions

The goal of achieving validly labeled data to differentiate between clean and polluted images by verified content becomes clearly defined. Once the valid labels are acquired, the current approach can be repeated so that Grad-CAM and SHAP will become valid diagnostic approaches. Further investigation might involve refining later layers of MobileNetV2 architecture and testing deeper networks like ResNet50 and EfficientNet [references: He et al., 2016; Tan et al., 2019]. Enlarging the database by incorporating variation within lighting, types of water, and different cameras will help avoid overfitting. Robustness testing with regard to blur, noise, and illumination variability will contribute to assessing field readiness of the tool. Repeated analysis with calculation of confidence intervals and testing on a separate dataset will add to statistical validity of the results. In the long run, the model might assist in inexpensive preliminary screening, highlighting questionable areas to check in a laboratory.

6. Conclusion

For the current work, there was a need to determine if a lighter CNN could classify water images as either being polluted or not on a publicly available dataset and compare the results with MobileNetV2 transfer learning model. The custom network showed good ability to learn from the training data since it achieved 79.06% training accuracy but had poor generalization ability since its validation accuracy stood at 43.10%, implying overfitting. The MobileNetV2 model with a frozen backbone was slightly better when generalizing, having 50.00% validation accuracy. As expected, this is due to the transferred ImageNet features. However, the critical finding for this project came out of the audit of the data pipeline where it became evident that the two classes were created through the random splitting of unlabelled data. This is the reason why the performance was close to random chance and not anything to do with architecture as initially thought. Therefore, this is a data quality problem rather than an architectural issue. Even though both SHAP and Grad-CAM were mentioned as possible forms of interpretation, they were rightfully put aside considering the inappropriate interpretation in the absence of actual labeling. In practice, it is now evident that the main limitation on this type of task is the proper labeling and number of data as opposed to choosing models. With well-labeled larger datasets and properly tuned transfer learning algorithm, similar results as those obtained in the controlled water imaging scenarios can be attained.

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